

Group assignment one for the course Food Hygiene and safety

### **Group 1. Types and Roles of Microorganisms in Food**

- **Objective:** Understand different types of microorganisms in food and their roles.
- **Tasks:**
  - Classify bacteria, yeasts, molds, and viruses found in food.
  - Describe their roles in food spoilage, foodborne illness, and fermentation.
  - Present case studies on common microorganisms in various foods.

### **Group 2. Pathogenic vs. Beneficial Microorganisms**

- **Objective:** Distinguish between harmful and beneficial microorganisms.
- **Tasks:**
  - Identify common foodborne pathogens and their health impacts.
  - Explore beneficial microorganisms used in food production.
  - Discuss methods to control pathogens while promoting beneficial microbes.

### **Group 3. Environmental Factors Influencing Microbial Growth**

- **Objective:** Examine factors like pH, temperature, moisture, and oxygen on microbial growth.
- **Tasks:**
  - Investigate how these factors affect microbial growth rates.
  - Conduct an experiment on how one factor impacts microbial growth.
  - Suggest control methods to reduce spoilage.

### **Group 4. Foodborne Pathogens and Prevention Strategies**

- **Objective:** Study major foodborne pathogens and strategies to prevent contamination.
- **Tasks:**
  - Identify major pathogens (e.g., *Salmonella*, *E. coli*, *Listeria*).
  - Describe their growth characteristics and common food sources.
  - Research effective contamination prevention methods.

### **Group 5. Food Preservation Techniques and Their Microbial Implications**

- **Objective:** Review food preservation methods and their effects on microorganisms.
- **Tasks:**
  - Explain techniques like pasteurization, fermentation, and freezing.
  - Discuss how each method impacts microbial survival.
  - Propose a suitable preservation method for a specific food, justifying it based on microbial considerations.

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